UNIT 2 :-

Exercise 2.1

Question : 1

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 7 & -1 \\ -3 & 1 & 4 \end{bmatrix} \qquad \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}$$

$$\rightarrow M_{11} = \begin{vmatrix} 7 & -1 \\ 1 & 4 \end{vmatrix}$$

$$= 29$$

$$C_{11} = (-1)^{1+1} M_{11} \qquad (-1)^2$$

$$= 1 \cdot 29$$

$$= 29$$

$$\rightarrow M_{12} = \begin{vmatrix} 6 & -1 \\ -3 & 4 \end{vmatrix}$$

$$= 21$$

$$C_{12} = (-1)^{1+2} M_{12}$$

$$= -1 \cdot 21$$

$$= -21$$

→ $M_{13} = \begin{vmatrix} 6 & 7 \\ -3 & 1 \end{vmatrix}$

$= 27$

$C_{13} = (-1)^{1+3} M_{13}$

$= 1 \cdot 27$

$= 27$

→ $M_{21} = \begin{vmatrix} -2 & 3 \\ 1 & 4 \end{vmatrix}$

$= -11$

$C_{21} = (-1)^{2+1} M_{21}$

$= -1 \cdot (-11)$

$= 11$

→ $M_{22} = \begin{vmatrix} 1 & 3 \\ -3 & 4 \end{vmatrix}$

$= 13$

$C_{22} = (-1)^{2+2} \cdot M_{22}$

$= 1 \cdot 13$

$= 13$

→ $M_{23} = \begin{vmatrix} 1 & -2 \\ -3 & 1 \end{vmatrix}$

$= -5$

$$C_{23} = (-1)^{2+3} M_{23}$$

$$= 5$$

$$\rightarrow M_{31} = \begin{vmatrix} -2 & 3 \\ 7 & -1 \end{vmatrix}$$

$$= -19$$

$$C_{31} = (-1)^{3+1} \cdot M_{31}$$

$$= -19$$

$$\rightarrow M_{32} = \begin{vmatrix} 1 & 3 \\ 6 & -1 \end{vmatrix}$$

$$= -19$$

$$C_{32} = (-1)^{3+2} M_{32}$$

$$= 19$$

$$\rightarrow M_{33} = \begin{vmatrix} 1 & -2 \\ 6 & 7 \end{vmatrix}$$

$$= 19$$

$$C_{33} = (-1)^{3+3} \cdot M_{33}$$

$$= 19$$

Q2:—

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 3 & 6 \\ 0 & 1 & 4 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} 3 & 6 \\ 1 & 4 \end{vmatrix}$$

$$= 6$$

$$C_{11} = (-1)^{1+1} \cdot M_{11}$$

$$= 6$$

$$M_{12} = \begin{vmatrix} 3 & 6 \\ 0 & 4 \end{vmatrix}$$

$$= 12$$

$$C_{12} = (-1)^{1+2} M_{12}$$

$$= -12$$

$$M_{13} = \begin{vmatrix} 3 & 3 \\ 0 & 1 \end{vmatrix}$$

$$= 3$$

$$C_{13} = (-1)^{1+3} \cdot M_{13}$$

$$= 3$$

$$M_{21} = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 2$$

$$C_{21} = (-1)^{2+1} \cdot M_{21}$$

$$= -2$$

$$\rightarrow M_{22} = \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} = 4$$

$$C_{22} = (-1)^{2+2} \cdot M_{22}$$

$$= 4$$

$$\rightarrow M_{23} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

$$C_{23} = (-1)^{2+3} \cdot M_{23}$$

$$= -1$$

$$\rightarrow M_{31} = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = 0$$

$$C_{31} = (-1)^{3+1} \cdot M_{31}$$

$$= 0$$

$$\rightarrow M_{32} = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = 0$$

$$C_{32} = (-1)^{3+2} \cdot M_{32}$$

$$= 0$$

$$\rightarrow M_{33} = \begin{vmatrix} 1 & 1 \\ 3 & 3 \end{vmatrix} = 0$$

$$C_{21} = (-1)^{2+1} \cdot M_{21}$$

$$= -2$$

$$\rightarrow M_{22} = \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} = 4$$

$$C_{22} = (-1)^{2+2} \cdot M_{22}$$

$$= 4$$

$$\rightarrow M_{23} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

$$C_{23} = (-1)^{2+3} \cdot M_{23}$$

$$= -1$$

$$\rightarrow M_{31} = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = 0$$

$$C_{31} = (-1)^{3+1} \cdot M_{31}$$

$$= 0$$

$$\rightarrow M_{32} = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = 0$$

$$C_{32} = (-1)^{3+2} \cdot M_{32}$$

$$= 0$$

$$\rightarrow M_{33} = \begin{vmatrix} 1 & 1 \\ 3 & 3 \end{vmatrix} = 0$$

$C_{33} = (-1)^{3+3} M_{33}$

$= 0$

Q3:—

(a-) $M_{13} = \begin{vmatrix} 0 & 0 & 3 \\ 4 & 1 & 14 \\ 4 & 1 & 2 \end{vmatrix}$

$= 3 \begin{vmatrix} 4 & 1 \\ 4 & 1 \end{vmatrix} = 0$

$C_{13} = (-1)^{1+3} \cdot M_{13}$

$= 0$

(b-) $M_{23} = \begin{vmatrix} 4 & -1 & 6 \\ 4 & 1 & 14 \\ 4 & 1 & 2 \end{vmatrix}$

$= 6 \begin{vmatrix} 4 & -1 \\ 4 & 1 \end{vmatrix} + 4 \begin{vmatrix} 1 & 14 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 4 & 14 \\ 4 & 2 \end{vmatrix}$

$= 48 - 48$

$= -96$

$C_{23} = (-1)^{2+3} \cdot C_{23}$

$= 96$

(c-) $M_{22} = \begin{vmatrix} 4 & 1 & 6 \\ 4 & 0 & 14 \\ 4 & 3 & 2 \end{vmatrix}$

$= 4\begin{vmatrix} 0 & 14 \\ 3 & 2 \end{vmatrix} - 1\begin{vmatrix} 4 & 14 \\ 4 & 2 \end{vmatrix} + 6\begin{vmatrix} 4 & 0 \\ 4 & 3 \end{vmatrix}$

$= -168 + 48 + 72$

$= -48$

$C_{22} = (-1)^{2+2} \cdot M_{22}$

$= -48$

(d-)

$M_{21} = \begin{vmatrix} -1 & 1 & 6 \\ 1 & 0 & 14 \\ 1 & 3 & 2 \end{vmatrix}$

$= -1\begin{vmatrix} 0 & 14 \\ 3 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & 14 \\ 1 & 2 \end{vmatrix} + 6\begin{vmatrix} 1 & 0 \\ 1 & 3 \end{vmatrix}$

$= 72$

$C_{21} = (-1)^{2+1} M_{21}$

$= -72$

Q4:-

$$A = \begin{bmatrix} 2 & 3 & -1 & 1 \\ -3 & 2 & 0 & 3 \\ 3 & -2 & 1 & 0 \\ 3 & -2 & 1 & 4 \end{bmatrix}$$

(a-) $M_{32} = \begin{vmatrix} 2 & -1 & 1 \\ -3 & 0 & 3 \\ 3 & 1 & 4 \end{vmatrix} = 2\begin{vmatrix} 0 & 3 \\ 1 & 4 \end{vmatrix} + 1\begin{vmatrix} -3 & 3 \\ 3 & 4 \end{vmatrix} + 1\begin{vmatrix} -3 & 0 \\ 3 & 1 \end{vmatrix}$

$= 2(-3) + (12 - 9) + 1(-3)$

$= -30$

$C_{32} = (-1)^{3+2} M_{32}$

$= 30$

(b-) $M_{44} = \begin{vmatrix} 2 & 3 & -1 \\ -3 & 2 & 0 \\ 3 & -2 & 1 \end{vmatrix} = 2\begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix} - 3\begin{vmatrix} 3 & 0 \\ 3 & 1 \end{vmatrix} - 1\begin{vmatrix} -3 & 2 \\ 3 & -2 \end{vmatrix}$

$= 13$

$C_{44} = (-1)^{4+4} 13 = 13$

(c-) $M_{41} = \begin{vmatrix} 3 & -1 & 1 \\ 2 & 0 & 3 \\ -2 & 1 & 0 \end{vmatrix}$

$= -3\begin{vmatrix} 0 & 3 \\ 1 & 0 \end{vmatrix} + 1\begin{vmatrix} 2 & 3 \\ -2 & 0 \end{vmatrix} + 1\begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix}$

$= -1$

$C_{41} = (-1)^{4+1} \cdot M_{41} = \ominus 1$

$$M_{24} = \begin{vmatrix} 2 & 3 & -1 \\ 3 & -2 & 1 \\ 3 & -2 & 1 \end{vmatrix}$$

(d-1)

$$= 2\begin{vmatrix} -2 & 1 \\ -2 & 1 \end{vmatrix} - 3\begin{vmatrix} 3 & 1 \\ 3 & 1 \end{vmatrix} - 1\begin{vmatrix} 3 & -2 \\ 3 & -2 \end{vmatrix}$$

$$= 0$$

$$C_{24} = (-1)^{2+4} \cdot M_{24}$$

$$= 0$$

Question : 5

let $A = \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}$

$$\text{Adj } A = \begin{bmatrix} 4 & -5 \\ 2 & 3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 5 \\ -2 & 4 \end{vmatrix}$$

$$= 12 + 10$$

$$= 22$$

$$A^{-1} = \frac{1}{|A|} \text{Adj } A$$

$$= \frac{1}{22} \begin{bmatrix} 4 & -5 \\ +2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2/11 & -5/22 \\ 1/11 & 3/22 \end{bmatrix}$$

Q6:—

let $A = \begin{bmatrix} 4 & 1 \\ 8 & 2 \end{bmatrix}$ $\quad Adj\ A = \begin{bmatrix} 2 & -1 \\ -8 & 4 \end{bmatrix}$

$$|A| = \begin{vmatrix} 4 & 1 \\ 8 & 2 \end{vmatrix}$$

$= 0$ not invertible.

A^{-1} does not exist.

Q7:—

let $A = \begin{bmatrix} -5 & 7 \\ -7 & -2 \end{bmatrix}$, $Adj\ A = \begin{bmatrix} -2 & -7 \\ 7 & -5 \end{bmatrix}$

$$|A| = \begin{vmatrix} -5 & 7 \\ -7 & -2 \end{vmatrix} = 59$$

$$A^{-1} = \frac{1}{|A|} Adj\ A$$

$$= \frac{1}{59} \begin{bmatrix} -2 & -7 \\ 7 & -5 \end{bmatrix} = \begin{bmatrix} -2/59 & -7/59 \\ 7/59 & -5/59 \end{bmatrix}$$

Q8:—

$$A = \begin{bmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{bmatrix} \quad \text{Adj } A = \begin{bmatrix} \sqrt{3} & -\sqrt{6} \\ -4 & \sqrt{2} \end{bmatrix}$$

$$|A| = \begin{vmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{vmatrix}$$

$$= \sqrt{6} - 4\sqrt{6}$$

$$= -3\sqrt{6}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj } A$$

$$= -\frac{1}{3\sqrt{6}} \begin{bmatrix} \sqrt{3} & -\sqrt{6} \\ -4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1/3\sqrt{2} & 1/3 \\ -4/3\sqrt{6} & -1/3\sqrt{3} \end{bmatrix}$$

Q9:—

let $A = \begin{bmatrix} a-3 & 5 \\ -3 & a-2 \end{bmatrix}$

$$|A| = \begin{vmatrix} a-3 & 5 \\ -3 & a-2 \end{vmatrix}$$

$= (a-3)(a-2) + 15$

$= a^2 - 5a + 22$

Q10:—

$$|A| = \begin{vmatrix} -2 & 7 & 6 \\ 5 & 1 & -2 \\ 3 & 8 & 4 \end{vmatrix} \begin{matrix} -2 & 7 \\ 5 & 1 \\ 3 & 8 \end{matrix}$$

$= [-8 - 42 + 240] - [18 + 32 + 140]$

$= 190 - 190$

$= 0$

Q11:—

$$\begin{vmatrix} -2 & 1 & 4 \\ 3 & 5 & -7 \\ 1 & 6 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & 1 & 4 \\ 3 & 5 & -7 \\ 1 & 6 & 2 \end{vmatrix} \begin{matrix} -2 & 1 \\ 3 & 5 \\ 1 & 6 \end{matrix}$$

$= [-20 - 7 + 72] - [20 + 84 + 6]$

$= 45 - 110 = -65$

Q12:—

$$|A| = \begin{vmatrix} -1 & 1 & -1 \\ 3 & 0 & 3 \\ 1 & 7 & 2 \end{vmatrix}$$

$$= [0 - 5 + 42] - [0 + 35 + 6]$$

$$= 37 - 41$$

$$= -4$$

Q13:—

$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{vmatrix} \begin{matrix} 3 & 0 \\ 2 & -1 \\ 1 & 9 \end{matrix}$$

$$= [12 + 0 + 0] - [0 + 135 + 0]$$

$$= -123$$

Q14:—

Q12:—

$$|A| = \begin{vmatrix} -1 & 1 & 2 & -1 & 1 \\ 3 & 0 & 5 & 3 & 0 \\ 1 & 7 & 2 & 1 & 7 \end{vmatrix}$$

$$= [0 - 5 + 42] - [0 + 35 + 6]$$

$$= 37 - 41$$

$$= -4$$

Q13:—

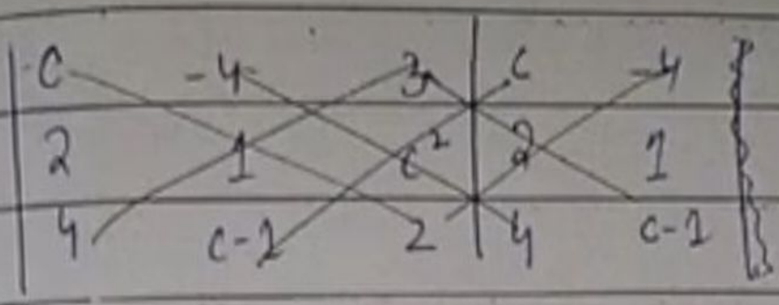
$$\begin{vmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 0 & 0 & 3 & 0 \\ 2 & -1 & 5 & 2 & -1 \\ 1 & 9 & -4 & 1 & 9 \end{vmatrix}$$

$$= [12 + 0 + 0] - [0 + 135 + 0]$$

$$= -123$$

Q14:—



$$= [2c - 16c^2 + 6c - 6] - [12 - 16 + c^4 - c^3]$$

$$= -16c^2 + 8c - 6 + 4 - c^4 + c^3$$

$$= -c^4 + c^2 - 16c^2 + 8c - 2$$

Q15:—

$$|A| = \begin{vmatrix} \lambda - 2 & 1 \\ -5 & \lambda + 4 \end{vmatrix} = 0$$

$$(\lambda - 2)(\lambda + 4) + 5 = 0$$

$$\lambda^2 + 2\lambda - 8 + 5 = 0$$

$$\lambda^2 + 2\lambda - 3 = 0$$

$$\lambda^2 + 3\lambda - \lambda - 3 = 0$$

$$\lambda(\lambda + 3) - 1(\lambda + 3) = 0$$

$\lambda + 3 = 0$ $\qquad$ $\lambda - 1 = 0$

$\lambda = -3$ $\qquad$ $\lambda = 1$

Q16:—

$$|A| = \begin{vmatrix} \lambda-4 & 0 & 0 \\ 0 & \lambda & 2 \\ 0 & 3 & \lambda-1 \end{vmatrix} = 0$$

$$(\lambda-4)\begin{vmatrix} \lambda & 2 \\ 3 & \lambda-1 \end{vmatrix} = 0$$

$$(\lambda-4)(\lambda^2-\lambda-6) = 0$$

$\boxed{\lambda = 4}$

$$\lambda^2-3\lambda+2\lambda-6 = 0$$

$$\lambda(\lambda-3)+2(\lambda-3) = 0$$

$$(\lambda-3)(\lambda+2) = 0$$

$\boxed{\lambda = 3}$ $\boxed{\lambda = -2}$

Q17:—

$$|A| = \begin{vmatrix} \lambda-1 & 0 \\ 2 & \lambda+2 \end{vmatrix} = 0$$

$$(\lambda-1)(\lambda+1) = 0$$

$$\lambda^2-1 = 0$$

$$\lambda = \pm 1$$

Q18:—

$$|A| = \begin{vmatrix} \lambda-4 & 4 & 0 \\ -1 & \lambda & 0 \\ 0 & 0 & \lambda-5 \end{vmatrix} = 0$$

$$(\lambda-5)\begin{vmatrix}\lambda-4 & 4\\ -1 & \lambda\end{vmatrix}=0$$

$$(\lambda-5)(\lambda^2-4\lambda+4)=0$$
$$(\lambda-5)(\lambda^2-2\lambda-2\lambda+4)=0$$
$$(\lambda-5)(\lambda(\lambda-2)-2(\lambda-2))=0$$
$$(\lambda-5)(\lambda-2)(\lambda-2)=0$$

$\lambda-5=0$ $\quad (\lambda-2)^2=0$

$\lambda=5$ $\quad \lambda=2$

Q19:—

$$\begin{vmatrix}3 & 0 & 0\\ 2 & -1 & 5\\ 1 & 9 & -4\end{vmatrix}$$

✓

By R_1 (a-) $|A|=3\begin{vmatrix}-1 & 5\\ 9 & -4\end{vmatrix}=3(4-45)=-123$

By C_1 (b-) $=3\begin{vmatrix}-1 & 5\\ 9 & -4\end{vmatrix}-2\begin{vmatrix}0 & 0\\ 9 & -4\end{vmatrix}+1\begin{vmatrix}0 & 0\\ -1 & 5\end{vmatrix}$

$=-123$

By R_2 (c-) $-2\begin{vmatrix}0 & 0\\ 9 & -4\end{vmatrix}-1\begin{vmatrix}3 & 0\\ 1 & -4\end{vmatrix}-5\begin{vmatrix}3 & 0\\ 1 & 9\end{vmatrix}$

$$\begin{vmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{vmatrix}$$

R_1 (a-) $= -1\begin{vmatrix} 0 & -5 \\ 7 & 2 \end{vmatrix} - 1\begin{vmatrix} 3 & -5 \\ 1 & 2 \end{vmatrix} + 2\begin{vmatrix} 3 & 0 \\ 1 & 7 \end{vmatrix}$

$= -1(0+35) - 1(6+5) + 2(21-0)$

$= -35 - 11 + 42$

$= -4$

R_2 (b-) $-3\begin{vmatrix} 1 & 2 \\ 7 & 2 \end{vmatrix} + 5\begin{vmatrix} -1 & 1 \\ 1 & 7 \end{vmatrix}$

$= 36 - 40$

$= -4$

R_3 (c-) $1\begin{vmatrix} 1 & 2 \\ 0 & -5 \end{vmatrix} - 7\begin{vmatrix} -1 & 2 \\ 3 & -5 \end{vmatrix} + 2\begin{vmatrix} -1 & 1 \\ 3 & 0 \end{vmatrix}$

$= 1(-5) - 7(5-6) + 2(-3)$

$= -5 + 7 - 6$

$= -4$

C_1 (d-) $-1\begin{vmatrix} 0 & -5 \\ 7 & 2 \end{vmatrix} - 3\begin{vmatrix} 1 & 2 \\ 7 & 2 \end{vmatrix} + 1\begin{vmatrix} 1 & 2 \\ 0 & -5 \end{vmatrix}$

$= -4$

C_2 (e-) $-1 \begin{vmatrix} 3 & -5 \\ 1 & 2 \end{vmatrix} - 7 \begin{vmatrix} -1 & 2 \\ 3 & -5 \end{vmatrix}$

$= -4$

C_3 (f-) $2 \begin{vmatrix} 3 & 0 \\ 1 & 7 \end{vmatrix} + 5 \begin{vmatrix} -1 & 1 \\ 1 & 7 \end{vmatrix} + 2 \begin{vmatrix} -1 & 1 \\ 3 & 0 \end{vmatrix}$

$= 42 - 40 + 6$

$= 4$

Question : 21

$$|A| = \begin{vmatrix} -3 & 0 & 7 \\ 2 & 5 & 1 \\ -1 & 0 & 5 \end{vmatrix}$$

Expand by C_2

$= 5 \begin{vmatrix} -3 & 7 \\ 1 & 5 \end{vmatrix}$

$= 5(-15 + 7)$

$= 5(-8)$

$= -40$

Question : 22

$$|A| = \begin{vmatrix} 3 & 3 & 1 \\ 1 & 0 & -4 \\ 1 & -3 & 5 \end{vmatrix}$$

Expand by R_2

$$= -1 \begin{vmatrix} 3 & 1 \\ -3 & 5 \end{vmatrix} + 4 \begin{vmatrix} 3 & 3 \\ 1 & -3 \end{vmatrix}$$

$$= -1(15 - 3) + 4(-9 - 3)$$

$$= -12 - 48$$

$$= -60$$

Question : 23

$$\begin{vmatrix} 1 & k & k^4 \\ 1 & k & k^2 \\ 1 & k & k^2 \end{vmatrix}$$

$$= 1 \begin{vmatrix} k & k^4 \\ k & k^2 \end{vmatrix} - k \begin{vmatrix} 1 & k^4 \\ 1 & k^2 \end{vmatrix} + k^2 \begin{vmatrix} 1 & k \\ 1 & k \end{vmatrix}$$

$$= 1(0) - k(0) + k^2(0)$$

$$= 0$$

Question : 24

$$|A| = \begin{vmatrix} k+1 & k-1 & 7 \\ 2 & k-3 & 4 \\ 5 & k+1 & k \end{vmatrix}$$

$$= k+1 \begin{vmatrix} k-3 & 4 \\ k+1 & k \end{vmatrix} - (k-1) \begin{vmatrix} 2 & 4 \\ 5 & k \end{vmatrix} + 7 \begin{vmatrix} 2 & k-3 \\ 5 & k+1 \end{vmatrix}$$

$$= (k+1)(k^2-3k-4k-4) - (k-1)(2k-20) + 7(2k+2-5k-15)$$

$$= (k+1)(k^2-7k-4) - (2k^2-20k-2k+20) + 7(-3k-13)$$

$$= (k^3+7k^2-4k+k^2-7k-4) - (2k^2-22k+20) - 21k-91$$

$$= k^3-6k^2-11k-4-2k^2+22k-20-21k-91$$

$$= k^3-8k^2+11k-21k-24-91$$

$$= k^3-8k^2-10k-115$$

Question : 25

$$\begin{vmatrix} 3 & 3 & 0 & 5 \\ 2 & 2 & 0 & -2 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{vmatrix}$$

Expand by C_3

$$= -3 \begin{vmatrix} 3 & 3 & 5 \\ 2 & 2 & -2 \\ 2 & 10 & 2 \end{vmatrix} - 3 \begin{vmatrix} 3 & 3 & 5 \\ 2 & 2 & -2 \\ 4 & 1 & 0 \end{vmatrix}$$

$$= -3\left(5(20-4)+2(30-6)+2(6-6)\right) - 3\left(5(2-8)+2(3-12)\right)$$

$$= -3(80+48) - 3(-30-18)$$

$= -240$

Q 26 :—

$$\begin{vmatrix} 4 & 0 & 0 & 1 & 0 \\ 3 & 3 & 3 & -1 & 0 \\ 1 & 2 & 4 & 2 & 3 \\ 9 & 4 & 6 & 2 & 3 \\ 2 & 2 & 4 & 2 & 3 \end{vmatrix}$$

expand by R_1

$$= 4 \begin{vmatrix} 3 & 3 & -1 & 0 \\ 2 & 4 & 2 & 3 \\ 4 & 6 & 2 & 3 \\ 2 & 4 & 2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 3 & 3 & 3 & 0 \\ 1 & 2 & 4 & 3 \\ 9 & 4 & 6 & 3 \\ 2 & 2 & 4 & 3 \end{vmatrix}$$

In 1st determinant 2nd & 4 row are same which implies its determinant value is zero.

$$|A| = -1 \begin{vmatrix} 3 & 3 & 3 & 0 \\ 1 & 2 & 4 & 3 \\ 9 & 4 & 6 & 3 \\ 2 & 2 & 4 & 3 \end{vmatrix}$$

Expand by R_2

$$= -1\left(3\begin{vmatrix} 2 & 4 & 3 \\ 4 & 6 & 3 \\ 2 & 4 & 3 \end{vmatrix} - 3\begin{vmatrix} 1 & 4 & 3 \\ 9 & 6 & 3 \\ 2 & 4 & 3 \end{vmatrix} + 3\begin{vmatrix} 1 & 2 & 3 \\ 9 & 4 & 3 \\ 2 & 2 & 3 \end{vmatrix}\right)$$

$$= -1\left[3(0) - 3(-6) + 3(-6)\right]$$

$$= -1\left(18 - 18\right)$$

$$= 0$$

Question 27:-

$$1 \cdot (-1) \cdot 1$$

$$= -1$$

Question 28:-

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= 2(2(2))$$

$$= 8$$

Question 29:-

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 4 & 3 & 0 \\ 1 & 2 & 3 & 8 \end{bmatrix}$$

$$= 0(2(3(8)))$$

$$= 0$$

Question : 30

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

$$= 1 \begin{bmatrix} 2 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= 1(2(12))$$

$$= 24$$

Question : 31

Exercise 2.2

Question : 1

$$A = \begin{bmatrix} -2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$|A| = -8 - 3$$
$$= -11$$

$$A^T = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix}$$

$$= -8 - 3$$
$$= -11$$

$$\det(A) = \det(A^T)$$

Question : 2

$$A = \begin{bmatrix} -6 & 1 \\ 2 & -2 \end{bmatrix}$$

$$|A| = 12 - 2 = 10$$

$$A^T = \begin{bmatrix} -6 & 2 \\ 1 & -2 \end{bmatrix}$$

$$= 12 - 2 = 10$$

Question : 3

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \\ 5 & -3 & 6 \end{bmatrix} \quad A^T = \begin{bmatrix} 2 & 1 & 5 \\ -1 & 2 & -3 \\ 3 & 4 & 6 \end{bmatrix}$$

$$|A| = [2(12+12) + 1(6-20) + 3(-3-10)]$$

$$= 2(24) + (-14) + 3(-13)$$

$$= 48 - 14 - 39$$

$$= -5$$

$$|A^T| = [2(12+12) - 1(-6+9) + 5(-4-6)]$$

$$= 2(24) - 1(3) + 5(-10)$$

$$= 48 - 3 - 50$$

$$= -5$$

Question : 4

$$A = \begin{bmatrix} 4 & 2 & -1 \\ 0 & 2 & -3 \\ -1 & 1 & 5 \end{bmatrix} \quad A^T = \begin{bmatrix} 4 & 0 & -1 \\ 2 & 2 & 1 \\ -1 & -3 & 5 \end{bmatrix}$$

$$|A| = [4(10+3) - 1(-6+2)] \qquad |A^T| = [4(10+3) - 1(-6+2)]$$

$$= 56 \qquad = 56$$

Question : 5

$1(1(-5(1)) + 0 + 0 + 0$

$= -5$

Question : 6

$1(1) + 0 + 0$

1

Question : 7

$1(-1(1)) + 0 + 0 + 0$

$= -1$

Question : 8

$1\left(-\frac{1}{3}(1)\right) + 0 + 0 + 0$

$= -\frac{1}{3}$

Question : 9

$$\begin{vmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{vmatrix}$$

$$= R_1+R_2 \begin{vmatrix} 1 & 1 & 7 \\ 2 & 7 & -2 \\ 0 & 1 & 5 \end{vmatrix}$$

$$= R_2+2R_1 \begin{vmatrix} 1 & 1 & 7 \\ 0 & 9 & 12 \\ 0 & 1 & 5 \end{vmatrix}$$

Expand by C_1

$$= 1 \begin{vmatrix} 9 & 12 \\ 1 & 5 \end{vmatrix}$$

$$= 1\,(45-12)$$

$$= 33$$

Question : 10

$$= \begin{vmatrix} 3 & 6 & -9 \\ 0 & 0 & -2 \\ -2 & 1 & 5 \end{vmatrix}$$

$$= R_1+R_3 \begin{vmatrix} 1 & 7 & -4 \\ 0 & 0 & -2 \\ -2 & 1 & 5 \end{vmatrix}$$

$$= R_3+2R_1 \begin{vmatrix} 1 & 7 & -4 \\ 0 & 0 & -2 \\ 0 & 15 & -3 \end{vmatrix}$$

Expand by C_1

$$1 \begin{vmatrix} 0 & -2 \\ 15 & -3 \end{vmatrix}$$

$$= 30$$

Question : 11

$$= \begin{vmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix}$$

$$= \begin{matrix} R_1 - 2R_2 \end{matrix} \begin{vmatrix} 0 & 1 & 1 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix}$$

$$R_1 \leftrightarrow R_2 \quad (-1) \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix}$$

Expand by C_1

$$-1 \begin{vmatrix} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{vmatrix}$$

$$-1 \left[1(3-0) - 1(6-0) - 1(4-1) \right]$$

$-1[3 - 6 - 3]$

$= 6$

Question 12:—

$$\begin{vmatrix} 1 & -3 & 0 \\ -3 & 4 & 1 \\ 5 & -2 & 2 \end{vmatrix}$$

$$\begin{array}{l|} \\ R_2+2R_1 \\ R_3-5R_1 \end{array} \begin{vmatrix} 1 & -3 & 0 \\ 0 & -2 & 1 \\ 0 & 13 & 2 \end{vmatrix}$$

Expand by C_1

$= 1(-4 - 13)$

$= -17$

Question 13:—

$$\begin{vmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{vmatrix}$$

$$
\begin{vmatrix}
1 & 3 & 1 & 5 & 3 \\
0 & 1 & 2 & 6 & 8 \\
0 & 0 & 1 & 0 & 1 \\
0 & 0 & 2 & 1 & 1 \\
0 & 0 & 0 & 1 & 1
\end{vmatrix}
\quad R_2 + 2R_1
$$

Expand by C_1

$$
1 \begin{vmatrix}
-1 & 2 & 6 & 8 \\
0 & 1 & 0 & 1 \\
0 & 2 & 1 & 1 \\
0 & 0 & 1 & 1
\end{vmatrix}
$$

Expand by C_1

$$
-1 \begin{vmatrix}
1 & 0 & 1 \\
2 & 1 & 1 \\
0 & 1 & 1
\end{vmatrix}
$$

$$
\begin{vmatrix}
1 & 0 & 1 \\
0 & 1 & 1 \\
0 & 1 & 1
\end{vmatrix}
\quad R_2 - 2R_1
$$

(-1)

Expand by C_1

$$= -(1 \cdot (1 + 1))$$

$$= -1(2) = -2$$

Question 14:—

$$= \begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 9 & 6 & 1 \end{vmatrix}$$

$$\begin{matrix} R_2 - 5R_1 \\ = R_3 + R_1 \\ R_4 - 2R_1 \end{matrix} \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 12 & 0 & -1 \end{vmatrix}$$

Expand by C_1

$$= 1 \begin{vmatrix} 1 & -9 & -2 \\ 0 & -3 & -1 \\ 12 & 0 & -1 \end{vmatrix}$$

$$= R_3 - 12R_1 \begin{vmatrix} 1 & -9 & -2 \\ 0 & -3 & -1 \\ 0 & 108 & 23 \end{vmatrix}$$

Expand by C_1

$$= 1(-69 + 108)$$

$$= 39$$

Question 14:—

$$= \begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 9 & 6 & 1 \end{vmatrix}$$

$$\begin{matrix} R_2 - 5R_1 \\ = R_3 + R_1 \\ R_4 - 2R_1 \end{matrix} \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 12 & 0 & -1 \end{vmatrix}$$

Expand by C_1

$$= 1 \begin{vmatrix} 1 & -9 & -2 \\ 0 & -3 & -1 \\ 12 & 0 & -1 \end{vmatrix}$$

$$= R_3 - 12R_1 \begin{vmatrix} 1 & -9 & -2 \\ 0 & -3 & -1 \\ 0 & 108 & 23 \end{vmatrix}$$

Expand by C_1

$$= 1(-69 + 108)$$

$$= 39$$

Question : 15

As $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$

$R_1 \leftrightarrow R_2$

So, $(-1)\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix}$

$R_1 \leftrightarrow R_3$

$(-1)(+1)\begin{vmatrix} d & e & f \\ g & h & i \\ a & b & c \end{vmatrix} = -6$

$R_2 \leftrightarrow R_3$

Question : 16

As $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = (-6)$

$-\begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix} = 6$

$R_1 \leftrightarrow R_3$

Question : 17

As $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$ ⟩ $\begin{bmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{bmatrix}$

$$\begin{vmatrix} 3a & 3b & 3c \\ d & e & f \\ g & h & i \end{vmatrix} = -18$$

$$\begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ g & h & i \end{vmatrix} = 18$$

$$\begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix} = 72$$

Question : 18

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6 \qquad \begin{bmatrix} a+d & b+e & c+f \\ -d & -e & -f \\ g & h & i \end{bmatrix}$$

R_1+R_2 then $det(A) = det(B)$

$$\begin{vmatrix} a+d & b+e & c+f \\ -d & -e & -f \\ g & h & i \end{vmatrix} = 6$$

Question : 19

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6 \quad , \quad \begin{vmatrix} a+g & b+h & c+i \\ d & e & f \\ g & h & i \end{vmatrix}$$

$R_1 + R_3$

$$\begin{vmatrix} a+g & b+h & c+i \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

As $R_1 + R_3$ of first matrix so,

det (first matrix) = det (coming matrix)

Question : 20

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

$R_3 + 3R_2$

∴ Multiply ~~by~~ by 3 of R_1 is added to R_3 so,

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g+3a & h+3b & i+3c \end{vmatrix} = -6$$

det (A) = det (B)

$2(R_2)$

$$\begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ g+3a & h+3b & i+3c \end{vmatrix} = -12$$

As

Question : 21

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix} = -6$$

$$\begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix} = +18$$

Question : 22

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

As matrix R_3 is row of zeros to

$$\begin{vmatrix} a & b & c \\ d & e & f \\ 2a & 2b & 2c \end{vmatrix} = 0$$

det(A) = 6

$$2\begin{vmatrix} a & b & c \\ d & e & f \\ a & b & c \end{vmatrix} = 0$$

Question : 23

Let $|A| = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$$\overset{C_2-C_1}{=}{}_{C_3-C_1}\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix}$$

= Expand by R_1

$$\begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix}$$

$$= [(b-a)(c^2-a^2)] - [(c-a)(b^2-a^2)]$$
$$= [(b-a)(c-a)(c+a)] - [(c-a)(b-a)(b+a)]$$
$$= (b-a)(c-a)[(c+a)-(b+a)]$$
$$= (b-a)(c-a)(c+a-b-a)$$
$$= (b-a)(c-a)(c-b)$$

Question : 31

Exercise 2.3

Question : 1

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}, k = 2$$

So, $kA = \begin{bmatrix} -2 & 4 \\ 6 & 8 \end{bmatrix}$

$\det(kA) = k^n \det(A)$ $\quad n = $ no. of rows

So, $= (-2)(8) - (6)(4) = 2^2\{(-1)(4) - (3)(2)\}$

$= 4[-4 - 6]$

$-16 - 24 = 4[-10]$

$-40 = -40$

L.H.S = R.H.S

Hence, proved.

Question : 2

$A = \begin{bmatrix} 3 & 2 \\ 5 & -2 \end{bmatrix}, k = -4, n = 2$ $\quad kA = \begin{bmatrix} -8 & -8 \\ -20 & 8 \end{bmatrix}$

$\det A = -14$ $\quad \det(kA) = -64 - 160$

$= -224$

As $\det(kA) = k^n \det A$

$-224 = (-4)^2(-14)$

$-224 = (16)(-14)$

$-224 = -224$

Question : 3

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix} \qquad k = -2, \quad n = 3$$

$$kA = \begin{bmatrix} -4 & 2 & -6 \\ -6 & -4 & -2 \\ -2 & -8 & -10 \end{bmatrix}, \quad k^n = (-2)^3 = -8$$

$\det(kA) = k^n \det(A)$

$-4(24) - 2(56) - 6(40) = -8[2(6) + 14 + 3(10)]$

$-96 - 112 - 240 = -8(12 + 14 + 30)$

$-448 = -8(56)$

$-448 = -448$

Question : 4

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 1 & -2 \end{bmatrix} \Rightarrow k=3 \;,\; n=3$$

$$k^n = 3^3 = 27$$

$$kA = \begin{bmatrix} 3 & 3 & 3 \\ 0 & 6 & 9 \\ 0 & 3 & -6 \end{bmatrix}$$

$$\det(kA) = k^n \det(A)$$

$$3(-36-27) = 27(-4-3)$$

$$3(-63) = 27(-7)$$

$$-189 = -189$$

Question : 5

(a-) $\det(AB) = \det(BA)$

$$AB = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 1 & 1 & 2 \\ 5 & 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 9 & -1 & 8 \\ 31 & 1 & 17 \\ 10 & 0 & 2 \end{bmatrix}, \quad BA = \begin{bmatrix} -1 & -3 & 6 \\ 17 & 11 & 4 \\ 10 & 5 & 2 \end{bmatrix}$$

$$\det(AB) = 10(-25) + 2(70)$$

$$= -250 + 80$$

$$= -170$$

$$\det(BA) = -1(2) + 3(-6) + 6(-25)$$

$$= -2 - 18 - 150$$

$$= -170$$

So, $\det(AB) = \det(BA)$

(b-) $\det(A+B) = \det A + \det B$

$$A+B = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 1 & -1 & 3 \\ 7 & 1 & 2 \\ 5 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 3 \\ 10 & 4 & 2 \\ 5 & 0 & 3 \end{bmatrix}$$

$$\det(A+B) = +4 \begin{vmatrix} 3 & 3 \\ 5 & 3 \end{vmatrix} = +4(-6)$$

$$= -24.$$

$$\det A = 2 \begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} = 2(5) = 10$$

$$\det B = -1\begin{vmatrix} 7 & 2 \\ 5 & 1 \end{vmatrix} + 1\begin{vmatrix} 1 & 3 \\ 5 & 1 \end{vmatrix}$$

$$= -3 - 14$$

$$= -17$$

$$\det A + \det B = 10 - 17$$

$$= -7$$

$$\det (A+B) \neq \det A + \det B$$

$$-24 \neq -7$$

Question : 6

(a-) $\det (AB) = \det (BA)$

$$AB = \begin{bmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ -2 & 2 & 2 \end{bmatrix}\begin{bmatrix} 2 & -1 & -4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 15 & 26 \\ 2 & -4 & -3 \\ -2 & 10 & 12 \end{bmatrix}$$

$$\det (AB) = -108 + 160 - 118$$

$$= -66$$

$$(BA) = \begin{bmatrix} 2 & -1 & -4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ -2 & 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 8 & -3 \\ -6 & 14 & 7 \\ 5 & -2 & -5 \end{bmatrix}$$

$$\det(BA) = -280 - 276 + 490$$

$$= -66$$

So, $\det AB = \det BA$

$$-66 = -66$$

Question : 7

$$|A| = \begin{bmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{bmatrix}$$

$$= 5(-4+2) + 3(-2+5)$$

$$= 5(-2) + 3(3)$$

$$= -1$$

So, matrix is invertible.

Question : 8

$2(-12)+3(6)$

$-24+18$

-6

$$|A| = \begin{vmatrix} 2 & 0 & 3 \\ 0 & 3 & 2 \\ -2 & 0 & -4 \end{vmatrix}$$

$= +3(-8+6)$

$= +3(-2)$

$= \mp 6$

(Invertible)

Question : 9

$$|A| = \begin{vmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{vmatrix}$$

$= 2(2)$

$= 4$

Invertible

Question : 10

$$|A| = \begin{vmatrix} -3 & 0 & 1 \\ 5 & 0 & 6 \\ 8 & 0 & 3 \end{vmatrix}$$

As 5 is 0 so $|A| = 0$

(Not invertible)

Question : 11

$$|A| = \begin{vmatrix} 4 & 2 & 8 \\ -2 & 1 & -4 \\ 3 & 1 & 6 \end{vmatrix}$$

$$= 4(6+4) - 2(-12+12) + 8(-2-3)$$

$$= 4(10) - 2(0) + 8(-5)$$

$$= 40 - 0 - 40$$

$$= 0$$

(Not invertible)

Question : 12

$$|A| = \begin{vmatrix} 1 & 0 & -1 \\ 9 & -1 & 4 \\ 8 & 9 & -1 \end{vmatrix}$$

$$= -35 - 89$$

$$= -124 \quad (\text{Invertible})$$

Question : 13

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 8 & 1 & 0 \\ -5 & 3 & 6 \end{vmatrix}$$

$$= 2(6)$$

$$= 12$$

Invertible

Q14:—

$$|A| = \begin{vmatrix} \sqrt{2} & -\sqrt{7} & 0 \\ 3\sqrt{2} & -3\sqrt{7} & 0 \\ 5 & -9 & 0 \end{vmatrix}$$

C_3 is 0 so (not invertible)

Q15:—

$$|A| = \begin{vmatrix} k-3 & -2 \\ -2 & k-2 \end{vmatrix} = 0$$

$$(k-3)(k-2) - 4 = 0$$

$$k^2 - 5k + 6 - 4 = 0$$

$$k^2 - 5k + 2 = 0$$

$$\left(k - \frac{5-\sqrt{17}}{2}\right)\left(k - \frac{5+\sqrt{17}}{2}\right) = 0$$

So, for matrix to be invertible

$$k \neq \frac{5 \pm \sqrt{17}}{2}$$

Question: 16

$$|A| = \begin{vmatrix} k & 2 \\ 2 & k \end{vmatrix} = 0$$

$$k^2 - 4 = 0$$

$$(k-2)(k+2) = 0$$

So,

$k = 2$, $k = -2$

For matrix to be invertible $k \neq \pm 2$

Question : 17

$$|A| = 0$$

$$\begin{vmatrix} 1 & 2 & 4 \\ 3 & 1 & 6 \\ k & 3 & 2 \end{vmatrix} = 0$$

$$-16 + 24 + 8k = 0$$

$$8k + 8 = 0$$

$$k + 1 = 0$$

$$k = -1$$

For A^{-1} $k \neq -1$

Question : 18

$$|A| = 0 \Rightarrow \begin{vmatrix} 1 & 2 & 0 \\ k & 1 & k \\ 0 & 2 & 1 \end{vmatrix} = 0$$

$$1(1 - 2k) - 2(k) = 0$$

$$1 - 2k - 2k = 0$$

$$1 - 4k = 0$$

$$\frac{1}{4} = k$$

For A^{-1} $k \neq \frac{1}{4}$

Question : 19

(Adjoint Method)

$$|A| = \begin{vmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{vmatrix}$$

$$= 5(-4+2) + 3(-2+5)$$

$$= -5(-2) + 3(3)$$

$$= -10 + 9$$

$$\det A = -1$$

Adj A = Cof(A) — cofactor

$$C_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 0 \\ 4 & 8 \end{vmatrix} = -8$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} -1 & 0 \\ 2 & 3 \end{vmatrix} = +3$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} -1 & -1 \\ 2 & 4 \end{vmatrix} = -2$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 5 & 5 \\ 4 & 3 \end{vmatrix} = +5$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 5 \\ 2 & 3 \end{vmatrix} = -4$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 5 \\ 2 & 4 \end{vmatrix} = +2$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 5 & 5 \\ -1 & 0 \end{vmatrix} = 5$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 5 \\ -1 & 0 \end{vmatrix} = -5$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 2 & 5 \\ -1 & -1 \end{vmatrix} = 3$$

$$\text{Cof}(A) = \begin{bmatrix} -3 & 3 & -2 \\ 5 & -4 & 2 \\ 5 & -5 & 3 \end{bmatrix}$$

$$\text{Adj } A = (\text{Cof } A)^T$$

$$= \begin{bmatrix} -3 & 5 & 5 \\ 3 & -4 & -5 \\ -2 & 2 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj } A$$

$$= -\frac{1}{1} \begin{bmatrix} -3 & 5 & 5 \\ 3 & -4 & -5 \\ -2 & 2 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -5 & -5 \\ -3 & 4 & 5 \\ 2 & -2 & -3 \end{bmatrix}$$

Question : 20

$$|A| = \begin{vmatrix} 2 & 0 & 3 \\ 0 & 3 & 2 \\ -2 & 0 & -4 \end{vmatrix}$$

$$|A| = -6$$

$$A^{-1} = \frac{1}{\det(A)} (\text{Adj } A)$$

Adj A = ?

So,

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 2 \\ 0 & -4 \end{vmatrix} = -12$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 0 & 2 \\ -2 & -4 \end{vmatrix} = -4$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 3 \\ -2 & 0 \end{vmatrix} = 6$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 3 \\ 0 & -4 \end{vmatrix} = 0$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 3 \\ -2 & -4 \end{vmatrix} = -2$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 0 \\ -2 & 0 \end{vmatrix} = 0$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 3 \\ 3 & 2 \end{vmatrix} = -9$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} = -4$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6$$

$$\text{Adj } A = \begin{bmatrix} -12 & -4 & 6 \\ 0 & -2 & 0 \\ -9 & -4 & 6 \end{bmatrix}$$

$$\text{Adj } A = (C_{ij})^T$$

$$= \begin{bmatrix} -12 & 0 & -9 \\ -4 & -2 & -4 \\ 6 & 0 & 6 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-6} \begin{bmatrix} -12 & 0 & -9 \\ -4 & -2 & -4 \\ 6 & 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & -3/2 \\ 2/3 & 1/3 & 2/3 \\ -1 & 0 & -1 \end{bmatrix}$$

Question : 21

$$|A| = \begin{vmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{vmatrix}$$

$|A| = 4$

C of A $\Rightarrow$

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 1 & -3 \\ 0 & 2 \end{vmatrix} = 2$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 0 & -3 \\ 0 & 2 \end{vmatrix} = 0$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -3 & 5 \\ 0 & 2 \end{vmatrix} = 6$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 5 \\ 0 & 2 \end{vmatrix} = 4$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 2 & -3 \\ 0 & 0 \end{vmatrix} = 0$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -3 & 5 \\ 1 & -3 \end{vmatrix} = 4$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 5 \\ 0 & -3 \end{vmatrix} = 6$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 2 & -3 \\ 0 & 1 \end{vmatrix} = 2$$

Co. $A = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 4 & 0 \\ 4 & 6 & 2 \end{bmatrix}$

$$\text{Adj } A = (\text{Co. } A)^T = \begin{bmatrix} 2 & 6 & 4 \\ 0 & 4 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 6 & 4 \\ 0 & 4 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 3/2 & 1 \\ 0 & 1 & 3/2 \\ 0 & 0 & 1/2 \end{bmatrix}$$

Question : 22

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 8 & 1 & 0 \\ -5 & 3 & 6 \end{vmatrix}$$

$|A| = 12$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 5 \\ 0 & -3 \end{vmatrix} = 6$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 2 & -3 \\ 0 & 1 \end{vmatrix} = 2$$

$$\text{Co}f\ A = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 4 & 0 \\ 4 & 6 & 2 \end{bmatrix}$$

$$\text{Adj } A = (\text{Co}f\ A)^T = \begin{bmatrix} 2 & 6 & 4 \\ 0 & 4 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 6 & 4 \\ 0 & 4 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 3/2 & 1 \\ 0 & 1 & 3/2 \\ 0 & 0 & 1/2 \end{bmatrix}$$

Question : 22

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 8 & 1 & 0 \\ -5 & 3 & 6 \end{vmatrix}$$

$$|A| = 12$$

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 1 & 0 \\ 3 & 6 \end{vmatrix} = 6$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 8 & 0 \\ -5 & 6 \end{vmatrix} = -48$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 8 & 1 \\ -5 & 3 \end{vmatrix} = 29$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 0 \\ 3 & 6 \end{vmatrix} = 0$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 0 \\ -5 & 6 \end{vmatrix} = 12$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 0 \\ -5 & 3 \end{vmatrix} = -6$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} = 0$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 0 \\ 8 & 0 \end{vmatrix} = 0$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 2 & 0 \\ 8 & 1 \end{vmatrix} = 2$$

$$\text{Cof}(A) = \begin{bmatrix} 6 & -48 & 29 \\ 0 & 12 & -6 \\ 0 & 0 & 2 \end{bmatrix}$$

Then, the adjoint matrix is given by

$$adj(A) = (Cof(A))^T = \begin{bmatrix} 6 & 0 & 0 \\ -48 & 12 & 0 \\ 29 & -6 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(Adj\,A)$$

$$A^{-1} = \frac{1}{12}\begin{bmatrix} 6 & 0 & 0 \\ -48 & 12 & 0 \\ 29 & -6 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 0 & 0 \\ -4 & 1 & 0 \\ 29/12 & -1/2 & 1/6 \end{bmatrix}$$

A is invertible

Question: 23

$$= \begin{vmatrix} 1 & 3 & 1 & 1 \\ 2 & 5 & 2 & 2 \\ 1 & 3 & 8 & 9 \\ 1 & 3 & 2 & 2 \end{vmatrix}$$

$$\begin{matrix} R_2-2R_1 \\ = R_3-R_1 \\ R_4-R_1 \end{matrix} \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 7 & 8 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1 & 3 & 1 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{vmatrix}$$

$R_3 - 6R_4$

Expand by R_1

$$1 \begin{vmatrix} -1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 1 & 1 \end{vmatrix}$$

Expand by R_1

$= 1(-1(1-2))$

$= -1(-1)$

$= +1$

Cof (A) for 4×4 matrix

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 2 & 2 \\ 3 & 8 & 9 \\ 3 & 2 & 2 \end{vmatrix} = -4$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 2 & 2 \\ 1 & 8 & 9 \\ 1 & 2 & 2 \end{vmatrix} = 2$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 2 & 5 & 2 \\ 1 & 3 & 9 \\ 1 & 3 & 2 \end{vmatrix} = -7$$

$$C_{14} = (-1)^{1+4} \begin{vmatrix} 2 & 5 & 2 \\ 1 & 3 & 8 \\ 1 & 3 & 2 \end{vmatrix} = 6$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 3 & 1 & 1 \\ 3 & 8 & 9 \\ 3 & 2 & 2 \end{vmatrix} = 3$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 1 & 1 \\ 1 & 8 & 9 \\ 1 & 2 & 2 \end{vmatrix} = -1$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 3 & 1 \\ 1 & 3 & 9 \\ 1 & 3 & 2 \end{vmatrix} = 6$$

$$C_{24} = (-1)^{2+4} \begin{vmatrix} 1 & 3 & 1 \\ 1 & 3 & 8 \\ 1 & 3 & 2 \end{vmatrix} = 0$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 3 & 1 & 1 \\ 5 & 2 & 2 \\ 3 & 2 & 2 \end{vmatrix} = 0$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 2 \end{vmatrix} = 0$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 3 & 2 \end{vmatrix} = -1$$

$$C_{34} = (-1)^{3+4} \begin{vmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 3 & 2 \end{vmatrix} = 1$$

$$C_{41} = (-1)^{4+1} \begin{vmatrix} 3 & 1 & 1 \\ 5 & 2 & 2 \\ 3 & 8 & 9 \end{vmatrix} = -1$$

$$C_{42} = (-1)^{4+2} \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 8 & 9 \end{vmatrix} = 0$$

$$C_{43} = (-1)^{4+3} \begin{vmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 3 & 9 \end{vmatrix} = -8$$

$$C_{44} = (-1)^{4+4} \begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 5 & 8 \end{vmatrix} = -7$$

$$\text{Cof}(A) = \begin{bmatrix} -4 & 2 & -7 & 6 \\ 3 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 0 & 8 & -7 \end{bmatrix}^T$$

$$\text{Adj } A = \begin{vmatrix} -4 & 3 & 0 & -1 \\ 2 & -1 & 0 & 0 \\ -7 & 0 & -1 & 8 \\ 6 & 0 & 1 & -7 \end{vmatrix}$$

$$A^{-1} = \begin{bmatrix} -4 & 3 & 0 & -1 \\ 2 & -1 & 0 & 0 \\ -7 & 0 & -1 & 8 \\ 6 & 0 & 1 & -7 \end{bmatrix}$$

Question : 24

$$7x_1 - 2x_2 = 3$$
$$3x_1 + x_2 = 5$$

$$A = \begin{bmatrix} 7 & -2 \\ 3 & 1 \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$b = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$|A| = 7 - (-6)$$
$$= 13$$

$$x_1 = \frac{\begin{vmatrix} 3 & -2 \\ 5 & 1 \end{vmatrix}}{|A|} = \frac{3+10}{13} = 1$$

$$x_2 = \frac{\begin{vmatrix} 7 & 3 \\ 3 & 5 \end{vmatrix}}{|A|} = \frac{35-9}{13} = 2$$

$x_1 = 1 \quad , x_2 = 2$

Question : 25

$$|A| = \begin{vmatrix} 4 & 5 & 0 \\ 11 & 1 & 2 \\ 1 & 5 & 2 \end{vmatrix}$$

$$= 4(2-10) - 5(22-2)$$

$$= 4(-8) - 5(20)$$

$$= -32 - 100$$

$$= -132$$

$$|A_x| = \begin{vmatrix} 2 & 5 & 0 \\ 3 & 1 & 2 \\ 1 & 5 & 2 \end{vmatrix}$$

$$= 2(2-10) - 5(6-2)$$

$$= 2(-8) - 5(4)$$

$$-16 - 20$$

$|A_x| = -36$

$$|A_y| = \begin{vmatrix} 4 & 2 & 0 \\ 11 & 3 & 2 \\ 1 & 1 & 2 \end{vmatrix}$$

$= 4(6-2) - 2(22-2)$

$= 4(4) - 2(20)$

$= 16 - 40$

$= -24$

$$|A_z| = \begin{vmatrix} 4 & 5 & 2 \\ 11 & 1 & 3 \\ 1 & 5 & 1 \end{vmatrix}$$

$= 4(1-15) - 5(11-3) + 2(55-1)$

$= 4(-14) - 5(8) + 2(54)$

$= -56 - 40 + 108$

$= -96 + 108$

$|A_z| = 12$

$x = \frac{|A_x|}{|A|} = \frac{+36}{+132} \frac{3}{11} = \frac{3}{11}$

$y = \frac{|A_y|}{|A|} = \frac{+24}{+132} \frac{2}{11} = \frac{2}{11}$

$z = \frac{-121}{132} = \frac{1}{11}$

Question :26

$$x - 4y + z = 6$$
$$4x - y + 2z = -1$$
$$2x + 2y - 3z = -20$$

$$|A| = \begin{bmatrix} 1 & -4 & 1 \\ 4 & -1 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

$$= 1(3-4) + 4(-12-4) + 1(8+2)$$
$$= -1 + 4(-16) + 1(10)$$
$$= -1 - 64 + 10$$
$$= -65 + 10$$
$$= -55$$

$$|Ax| = \begin{bmatrix} 6 & -4 & 1 \\ -1 & -1 & 2 \\ -20 & 2 & -3 \end{bmatrix}$$

$= 6(3-4) + 4(3+40) + 1(-2-20)$

$= 6(-1) + 4(43) + 1(-22)$

$= -6 + 172 - 22$

$= -6 + 150$

$= 144$

$$|A_y| = \begin{vmatrix} 1 & 6 & 1 \\ 4 & -1 & 2 \\ 2 & -20 & -3 \end{vmatrix}$$

$= -61$

$$|A_z| = \begin{vmatrix} 1 & -4 & 6 \\ 4 & -1 & -1 \\ 2 & 2 & -20 \end{vmatrix}$$

$= 230$

$x = \frac{|A_x|}{|A|}$ $\quad y = \frac{|A_y|}{|A|}$ $\quad z = \frac{|A_z|}{|A|}$

$= \frac{-144}{55}$ $\quad = \frac{-61}{55}$ $\quad = \frac{230}{55}$

Question : 27

$$x_1 - 3x_2 + x_3 = 4$$
$$2x_1 - x_2 = -2$$
$$4x_1 - 3x_3 = 0$$

$$|A| = \begin{vmatrix} 1 & -3 & 1 \\ 2 & -1 & 0 \\ 4 & 0 & -3 \end{vmatrix}$$

$$= 1(3) + 3(-6) + 1(4)$$
$$= 3 - 18 + 4$$
$$= -15 + 4$$
$$= -11$$

$$|Ax_1| = \begin{vmatrix} 4 & -3 & 1 \\ -2 & -1 & 0 \\ 0 & 0 & -3 \end{vmatrix} = +30$$

$$|Ax_2| = \begin{vmatrix} 1 & 4 & 1 \\ 2 & -2 & 0 \\ 4 & 0 & -3 \end{vmatrix} = 38$$

$$|Ax_3| = \begin{vmatrix} 1 & -3 & 4 \\ 2 & -1 & -2 \\ 4 & 0 & 0 \end{vmatrix} = 40$$

$x_1 = |Ax_1| / |A|$ $\quad x_2 = |Ax_2| / |A|$ $\quad x_3 = |Ax_3| / |A|$

$= \frac{-30}{11}$ $\quad = \frac{-38}{11}$ $\quad = \frac{-40}{11}$

$$|A_{x_3}| = \begin{vmatrix} -1 & -4 & -32 & 1 \\ 2 & -1 & 14 & 9 \\ -1 & 1 & 11 & 1 \\ 1 & -2 & -4 & -4 \end{vmatrix}$$

$$= -1269$$

$$|A_{x_4}| = \begin{vmatrix} -1 & -4 & 2 & -32 \\ 2 & -1 & 7 & 14 \\ -1 & 1 & 3 & 11 \\ 1 & -2 & 1 & -4 \end{vmatrix}$$

$$= 423$$

$$x_1 = \frac{|A_{x_1}|}{|A|} = \frac{-2115}{-423} = 5$$

$$x_2 = \frac{|A_{x_2}|}{|A|} = \frac{-3384}{-423} = 8$$

$$x_3 = \frac{|A_{x_3}|}{|A|} = \frac{-1269}{-423} = 3$$

$$x_4 = \frac{|A_{x_4}|}{|A|} = \frac{423}{-423} = -1$$

Question : 29

$$|A| = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 7 & -2 \\ 2 & 6 & -1 \end{bmatrix}$$

$= m$

$= 3(-7+12) + 1(1+4) + 1(-6-14)$

$= 3(5) + 1(5) + 1(-20)$

$= 15 + 5 - 20$

$= 20 - 20$

$= 0$

Question : 30

$$\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|A| = \cos^2\theta + \sin^2\theta = 1$$

$$C_{11} = (-1)^{1+1} \begin{vmatrix} \cos\theta & 0 \\ 0 & 1 \end{vmatrix} = \cos\theta$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} -\sin\theta & 0 \\ 0 & 1 \end{vmatrix} = \sin\theta$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} -\sin\theta & \cos\theta \\ 0 & 0 \end{vmatrix} = 0$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} \sin\theta & 0 \\ 0 & 1 \end{vmatrix} = -\sin\theta$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} \cos\theta & 0 \\ 0 & 1 \end{vmatrix} = \cos\theta$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} \cos\theta & \sin\theta \\ 0 & 0 \end{vmatrix} = 0$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} \sin\theta & 0 \\ \cos\theta & 0 \end{vmatrix} = 0$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} \cos\theta & 0 \\ -\sin\theta & 0 \end{vmatrix} = 0$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{vmatrix} = 1$$

Cof $A = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}^T$

Adj $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \checkmark$$

proved

Question : 31

$$A = \begin{bmatrix} 4 & 1 & 1 & 1 \\ 3 & 7 & -1 & 1 \\ 7 & 3 & -5 & 8 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

$R_2 - R_1$
$R_3 - 8R_1$
$R_4 - 2R_1$

$$\begin{bmatrix} 4 & 1 & 1 & 1 \\ -1 & 6 & -2 & 0 \\ -25 & -5 & -13 & 0 \\ -7 & -1 & -1 & 0 \end{bmatrix}$$

$$|A| = 1 \begin{vmatrix} -1 & 6 & -2 \\ -25 & -5 & -13 \\ -7 & -1 & -1 \end{vmatrix}$$

$$= -424$$

Ans =

$$|A_y| = 0$$

$$\boxed{y = 0}$$

Unit NO 3:—

EX 3.1

Q1:—

(a-)

$A=(1,5)$ $B=(4,1)$

$\vec{AB} = B-A = (4,1)-(1,5)$

$= (4-1, 1-5)$

$= (3,-4)$

x component is 3 & y component is -4.

(b-) Tail Head

$A=(2,3,0)$, $B=(0,0,4)$

$\vec{AB} = B-A = (0,0,4)-(2,3,0)$

$= (-2,-3,4)$

x component is -2 z component is 4

y component is -3

Q2:—

(a-)

$A=(2,3)$, $B=(-3,3)$

$\vec{AB} = B - A = (-3,3) - (2,3)$

$= (-3-2, 3-3)$

$= (-5, 0)$

x component is -5

y component is 0

(b-)

$A = (3,0,4)$, $B = (0,4,4)$

$\vec{AB} = B - A = (0,4,4) - (3,0,4)$

$= (-3, 4, 0)$

x component is -3

y component is 4

z component is 0

Q3:—

(a-) $\vec{P_1P_2} = P_2 - P_1 = (2,8) - (3,5)$

$= (-1, 3)$

x component is -1

y component is 3

(b-) $\vec{P_1P_2} = P_2 - P_1 = (2,4,2) - (5,-2,1)$

$= (-3, 6, 1)$

x component is -3., z component is 1.

y component is 6.

Q4:—

(a-) $\overrightarrow{P_1P_2} = P_2 - P_1 = (-4,-1) - (-6,2)$

$= (2,-3)$

x component is 2

y component is -3

(b-) $\overrightarrow{P_1P_2} = P_2 - P_1 = (-1,6,1) - (0,0,0)$

$= (-1,6,1)$

x component is -1

y component is 6

z component is 1

Q5:—

(a-) $A=(1,1)$, $B=(x,y)$, $\vec{u}=(1,2)$

$\overrightarrow{AB} = B - A = (x,y) - (1,1)$

$= (x-1, y-1)$

$\overrightarrow{AB} = \vec{u}$

$(x-1, y-1) = (1,2)$

$x - 1 = 1$ $\quad$ $y - 1 = 2$

$x = 2$ $\quad$ $y = 3$

$\boxed{B(2,3)}$

(b-) $u=(1,1,3)$, $B=(-1,-1,2)$

$A:(x,y,z)$

$\overrightarrow{AB} = B-A = (-1,-1,2)-(x,y,z)$

$= (-1-x, -1-y, 2-z)$

$\overrightarrow{AB} = \vec{u}$

$(-1-x, -1-y, 2-z) = (1,1,3)$

$-1-x = 1$ $\quad -1-y = 1$ $\quad 2-z = 3$

$-1-1 = x$ $\quad -1-1 = y$ $\quad 2-3 = z$

$\boxed{x = -2}$ $\quad \boxed{y = -2}$ $\quad \boxed{z = -1}$

$\boxed{A(-2,-2,-1)}$

Q6:—

(a-) $A(x,y)$ $\quad u=(1,2)$ $\quad B=(2,0)$

$\overrightarrow{AB} = B-A = (2,0)-(x,y)$

$= (2-x, 0-y)$

$= (2-x, -y)$

$\overrightarrow{AB} = \vec{u}$

$(2-x, -y) = (1,2)$

$2-x = 1$ $\quad -y = 2$

$1 = x$

$y = 2$

$A(1,-2)$

(b-) $B(x,y,z)$ $\quad A=(0,2,0)$, $u=(1,1,3)$

$\vec{AB} = B - A = (x,y,z) - (0,2,0)$

$= (x, y-2, z)$

$\vec{AB} = \vec{u}.$

$(x, y-2, z) = (1,1,3)$

$x = 1$ $\quad y-2=1$ $\quad z = 3$

$y = 3$

$(1,3,3)$

Q7:—

(a-) $P(P_1,P_2,P_3)$, $Q=(3,0,-5)$

$\vec{v} = (4,-2,-1)$

$\vec{u} = \vec{PQ} = Q - P = (3,0,-5) - (P_1,P_2,P_3)$

$= (3-P_1, -P_2, -5-P_3)$

$3-P_1 = 4$ $\quad -P_2 = -2$ $\quad -5-P_3 = -1$

$P_1 = -1$ $\quad P_2 = 2$ $\quad P_3 = -4$

$\boxed{P(-1, 2, -4)}$

(b-)

$P = (x, y, z)$

$\vec{u} = \vec{PQ} = Q - P = (3, 0, -5) - (x, y, z)$

$= (3-x, -y, -5-z)$

$\vec{u}$ is oppositely directed to $\vec{v} = (4, -2, 1)$

$3 - x = -4 \qquad -y = 2 \qquad -5 - z = 1$

$7 = x \qquad y = -2 \qquad z = -6$

$\boxed{P(7, -2, -6)}$

Q8:—

(a-) $Q = (x, y, z) \qquad u = v$

$\vec{u} = \vec{PQ} = Q - P = (x, y, z) - (-1, 3, -5)$

$= (x+1, y-3, z+5)$

$\vec{u}$ has the same direction as $\vec{v} = (6, 7, -3)$

$x+1=6$ $\quad y-3=7$ $\quad z+5=-3$

$x=5$ $\quad y=10$ $\quad z=-8$

$Q=(5,10,-8)$

(b-) $Q=(x,y,z)$

$\vec{u}=\vec{PQ}=Q-P=(x,y,z)-(-1,3,-5)$

$=(x+1,y-3,z+5)$

$\vec{u}$ is oppositely directed to $\vec{v}=(6,7,-3)$

$x+1=-6$ $\quad y-3=-7$ $\quad z+5=3$

$x=-7$ $\quad y=-4$ $\quad z=-2$

$Q=(-7,-4,-2)$

Q9:—

(a-)

$u+w=(4,-1)+(-3,-3)$

$=(1,-4)$

x - component is 1

y - component is -4

(b-)

$$v - 3u = (0,5) - 3(4,-1)$$
$$= (0,5) - (12,-3)$$
$$= (-12, 8)$$

x - component is -12

y - component is 8

(c-)

$$2(u - 5w) = 2[(4,-1) - 5(-3,-3)]$$
$$= 2[(4,-1) - (-15,-15)]$$
$$= 2[4-(-15), -1+15)$$
$$= 2(19, 14)$$
$$= (38, 28)$$

x - component is 38

y - component is 28

(d-)

$$3v - 2(u + 2w) = 3(0,5) - 2[(4,-1) + 2(-3,-3)]$$
$$= (0,15) - 2[(4,-1) + (-6,-6)]$$
$$= (0,15) - 2(-2,-7)$$
$$= (0,15) - (-4,-14)$$
$$= (4, 29)$$

x-component is 4
y-component is 29

Q10:—

(a-)

$v - w = (4, 0, -8) - (6, -1, -4)$

$= (-2, 1, -4)$

x-component is -2

y-component is 1

z-component is -4

(b-) $6u + 2v = 6(-3, 1, 2) + 2(4, 0, -8)$

$= (-18, 6, 12) + (8, 0, -16)$

$= (-10, 6, -4)$

(c-) $-3(v - 8w) = -3((4, 0, -8) - 8(6, -1, -4))$

$= -3((4, 0, -8) - (48, -8, -32))$

$= -3(-44, 8, 24)$

$= (132, -24, -72)$

(d-) $(2u - 7w) - (8v + u) =$

$[2(-3, 1, 2) - 7(6, -1, -4)] - [8(4, 0, -8) + (-3, 1, 2)]$

$= ((-6, 2, 4) - (42, -7, -28)) - ((32, 0, -64) + (-3, 1, 2))$

$= (-48, 9, 32) - (29, 1, -62)$

$= (-77, 8, 94)$

Q11:—

(a-)

$v - w = (4, 7, -3, 2) - (5, -2, 8, 1)$

$= (-1, 9, -11, 1)$

1st component is -1 , 2nd component is 9

3rd component is -11 , 4th component is 1

(b-)

$-u + (v - 4w) =$

$-(-3, 2, 1, 0) + ((4, 7, -3, 2) - 4(5, -2, 8, 1))$

$= (3, -2, -1, 0) + ((4, 7, -3, 2) - (20, -8, 32, 4))$

$= (3, -2, -1, 0) + (-16, 15, -35, -2)$

$= (-13, 13, -36, -2)$

(c-)

$6(u - 3v) = 6((-3, 2, 1, 0) - 3(4, 7, -3, 2))$

$= 6((-3, 2, 1, 0) - (12, 21, -9, 6))$

$= 6(-15, -19, 10, -6)$

$= (-90, -114, 60, -36)$

(d-)

$v-w)-(4u+v)=$

$(6(4,7,-3,2)-(5,-2,8,1))-(4(-3,2,1,0)+(4,7,-3,2))$

$=((24,42,-18,12)-(5,-2,8,1))-((-12,8,4,0)+(4,7,-3,2))$

$=(19,44,-26,11)-(-8,15,1,2)$

$=(27,29,-27,9)$

Q12:—

(a-)

$u+w=(0,4,-1,1,2)+(7,1,-4,-2,3)$

$=(7,5,-5,-1,5)$

(b-)

$3(2u-v)=3(2(1,2,-3,5,0)-(0,4,-1,1,2))$

$=3((2,4,-6,10,0)-(0,4,-1,1,2))$

$=3(2,0,-5,9,-2)$

$=(6,0,-15,27,-6)$

(c-)

$(3u-v)-(2u+4w)=$

$(3(1,2,-3,5,0)-(0,4,-1,1,2))-(2(1,2,-3,5,0)+4(7,1,-4,-2,3))$

$=3((3,6,-9,15,0)-(0,4,-1,1,2))-((2,4,-6,10,0)+(28,4,-16,-8,12))$

$=(3,2,-8,14,-2)-(30,8,-22,2,12)$

$= (-27, -6, 14, 12, -14)$

(d-)

$\frac{1}{2}(w - 5v + 2u) + v =$

$\frac{1}{2}\left((7,1,-4,-2,3) - 5(0,4,-1,1,2) + 2(1,2,-3,5,0)\right) + (0,4,-1,1,2)$

$\frac{1}{2}\left((7,1,-4,-2,3) - (0,20,-5,5,10) + (2,4,-6,10,0)\right) + (0,4,-1,1,2)$

$= \frac{1}{2}(9, -15, -5, 3, -7) + (0,4,-1,1,2)$

$= \left(\frac{9}{2}, \frac{-15}{2}, \frac{-5}{2}, \frac{3}{2}, \frac{-7}{2}\right) + (0,4,-1,1,2)$

$= \left(\frac{9}{2}, \frac{-7}{2}, \frac{-7}{2}, \frac{5}{2}, \frac{-3}{2}\right)$

$\frac{-15}{2} + 4$

$\frac{-15+8}{2} = \frac{-7}{2}$

Q13:—

$3u + v + 2w = 3x + 2w$

$\frac{3u + v - 2w - 2w}{3} = x$

$x = \frac{1}{3}\left[3u + v - 4w\right]$

$= \frac{1}{3}\left[3(-3,2,1,0) + (4,7,-3,2) - 4(5,-2,8,1)\right]$

$= \frac{1}{3}\left[(-9,6,3,0) + (4,7,-3,2) - (20,-8,32,4)\right]$

$$= \frac{1}{3}[-25, 21, -32, -2]$$

$$x = \left(\frac{-25}{3}, 7, \frac{-32}{3}, \frac{-2}{3}\right)$$

1st component is $\frac{-25}{3}$, 2nd component is 7
3rd component is $\frac{-32}{3}$, 4th component is $\frac{-2}{3}$

Q 14:—

$$2u - v + x = 7x + w$$

$$x = \frac{1}{6}[2u - v - w]$$

$$= \frac{1}{6}[2(1,2,-3,5,0) - (0,4,-1,1,2) - (7,1,-4,-2,3)]$$

$$= \frac{1}{6}[(2,4,-6,10,0) - (0,4,-1,1,2) - (7,1,-4,-2,3)]$$

$$= \frac{1}{6}(-5,-1,-1,11,-5)$$

$$= \left(\frac{-5}{6}, \frac{-1}{6}, \frac{-1}{6}, \frac{11}{6}, \frac{-5}{6}\right)$$

1st component is $-5/6$, 2nd component is $-1/6$
3rd component is $-1/6$, 4th component is $11/6$
5th component is $-5/6$

Q15:—

$u = s \cdot v$

(a-)

$(-2, 1, 0, 3, 5, 1) = s \cdot (4, 2, 0, 6, 10, 2) = a$

$-2 = 4s$ —— (1)

$1 = 2s$ —— (2)

$0 = 0$ —— (3)

$3 = 6s$ —— (4)

$5 = 10s$ —— (5)

$1 = 2s$ —— (6)

★ 1st (eq) we get $s = -1/2$ but from 2nd (eq) we get $s = 1/2$

★ $u = s \cdot a$ **Not parallel**

★ a is ~~not~~ parallel with u.

(b-)

$(-2, 1, 0, 3, 5, 1) = s \cdot (4, -2, 0, -6, -10, -2) = b$

$-2 = 4s$ —— (1)

$1 = -2s$ —— (2)

UNIT NO 5 :-

Exercise 5.1

Question : 1

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$A\vec{x} = \lambda\vec{x}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \lambda\vec{x}$$

$$\begin{bmatrix} 1-2 \\ 3-2 \end{bmatrix} = \lambda\vec{x}$$

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \lambda\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$-\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \lambda\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$Ax = -1x$

$\lambda = -1$

x is the Eigen vector of
-1 is the eigenvalue of A

Question : 2

$$A = \begin{bmatrix} 5 & -1 \\ 1 & 3 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$Ax = \lambda x = \begin{bmatrix} 5 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5-1 \\ 1+3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$Ax = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$Ax = 4x$$

x is the eigen vector of A
and eigen value is 4.

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Question = 3

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix} \qquad x = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4+0+1 \\ 2+6+2 \\ 1+0+4 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$$

$$= 5 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$Ax = 5x$$

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x is the eigenvector and

λ is the eigenvalue

Question : 4

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} ; x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-1-1 \\ -1+2-1 \\ -1-1+2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ax = 0 \cdot x$$

x is eigenvector of A and the eigenvalue is 0.

Question NO 5:-

(a-) $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

Take characteristics equation is

$\Rightarrow$

$$|A - \lambda I| = 0$$

$$A - \lambda I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix}$$

$$= (1-\lambda)(3-\lambda) - (8)$$

Date:

$|A - \lambda I| = 0$

$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$

$\lambda^2 - 4\lambda - 5 = 0$

$\lambda^2 - 5\lambda + \lambda - 5 = 0$

$\lambda(\lambda - 5) + 1(\lambda - 5) = 0$

$(\lambda - 5)(\lambda + 1) = 0$

$\lambda = 5 \qquad \lambda = -1$

- The characteristic equation is
  $\lambda^2 - 4\lambda - 5 = 0$
- The eigenvalues is
  $\lambda = 5 \;,\; \lambda = -1$

→ for $\lambda = 5$ put in $(A - \lambda I)$

$$= \begin{bmatrix} 1-5 & 4 \\ 2 & 3-5 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 4 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4x + 4y \\ -2x - 2y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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$-4x + 4y = 0$ — ①

$-2x - 2y = 0$ — ②

As eq ①

$4y = 4x$

$y = x$

As eq ②

$-2x = 2y$

$-x = y$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ x \end{bmatrix} = x\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is the basis for the eigenspaces to $\lambda = 5$

→

For $\lambda = -1$ put in $(A - \lambda I)$

$$= \begin{bmatrix} 1+1 & 4 \\ 2 & 3+1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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$2x + 4y = 0$

$2x + 4y = 0$

As eq (1)

$2x + 4y = 0$

$2x = -4y$

$x = -2y$

put $y = t$

$x = -2t$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2t \\ t \end{bmatrix}$$

$$= t \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ is the basis for eigenspace to $\lambda = -1$

(b-)

Date:

$$\begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} -2-\lambda & -7 \\ 1 & 2-\lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} -2-\lambda & -7 \\ 1 & 2-\lambda \end{vmatrix}$$

$$= (-2-\lambda)(2-\lambda) - (-7)$$

$$= -4 + 2\lambda - 2\lambda + \lambda^2 + 7$$

$$= \lambda^2 + 3$$ ✓

$$|A - \lambda I| = 0$$

$$\lambda^2 + 3 = 0$$

$$\lambda^2 = -3$$

$$\lambda = \pm\sqrt{3}\,i$$ ✓

★ The characteristic equation is $\lambda^2 + 3 = 0$

★ The Eigenvalues is $\lambda = +\sqrt{3}\,i$, $\lambda = -\sqrt{3}\,i$

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For $\lambda = \sqrt{3}i$ put in $(A-\lambda I)x = 0$

$$\begin{bmatrix} -2-\sqrt{3}i & -7 \\ 1 & 2-\sqrt{3}i \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(-2-\sqrt{3}i)x - 7y = 0 \quad \text{—(1)}$$

$$x + (2-\sqrt{3}i)y = 0 \quad \text{—(2)}$$

As eq (2)

$$x + (2-\sqrt{3}i)y = 0$$

$$x = -2+\sqrt{3}iy$$

put $y = t$

$$x = (-2+\sqrt{3}i)t$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (-2+\sqrt{3}i)t \\ t \end{bmatrix}$$

$$= t\begin{bmatrix} -2+\sqrt{3}i \\ 1 \end{bmatrix}$$

For $\lambda = -\sqrt{3}i$ put in $(A-\lambda I)x = 0$

$$\begin{bmatrix} -2+\sqrt{3}i & -7 \\ 1 & 2+\sqrt{3}i \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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$$(-2+\sqrt{3}i)x - 7y = 0 \quad \text{—①}$$

$$x + (2+\sqrt{3}i)y = 0 \quad \text{—②}$$

As eq ②

$$x = (-2-\sqrt{3}i)y$$

put $y = t$

$$x = (-2-\sqrt{3}i)t$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (-2-\sqrt{3}i)t \\ t \end{bmatrix}$$

$$= t \begin{bmatrix} -2-\sqrt{3}i \\ 1 \end{bmatrix}$$

★ $\begin{bmatrix} -2-\sqrt{3}i \\ 1 \end{bmatrix}$ is the basis for eigenspace to $\lambda = -\sqrt{3}i$

★ $\begin{bmatrix} -2+\sqrt{3}i \\ 1 \end{bmatrix}$ is the basis for eigenspace to $\lambda = \sqrt{3}i$

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(c-)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)(1-\lambda) - 0$$

$$= (1-\lambda)^2$$

$$|A - \lambda I| = 0$$

$$(1-\lambda)^2 = 0$$

$$(1-\lambda) = 0 \quad , \quad (1-\lambda) = 0$$

$$\boxed{1 = \lambda} \quad , \quad \boxed{1 = \lambda}$$

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The characteristic equation is $(1-\lambda)^2$

The eigenvalues is $\lambda=1$, $\lambda=1$

For $\lambda=1$ put in $(A-\lambda I)x = 0$

$$\begin{bmatrix} 1-1 & 0 \\ 0 & 1-1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

put $x=t$, $y=s$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ s \end{bmatrix} \quad \begin{bmatrix} t+0s \\ 0t+s \end{bmatrix}$$

$$= t\begin{bmatrix} 1 \\ 0 \end{bmatrix} + s\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

For $\lambda=1$, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is the basis for eigenspace.

(d-) $\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & -2 \\ 0 & 1-\lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & -2 \\ 0 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)(1-\lambda) - (0)$$

$$= (1-\lambda)^2$$

$$|A - \lambda I| = 0$$

$$(1-\lambda)^2 = 0$$

$$1 = \lambda$$

For $\lambda = 1$ put in $(A - \lambda I)x = 0$

$$\begin{bmatrix} 1-1 & -2 \\ 0 & 1-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-2y = 0$ put $x = t$

$\boxed{y = 0}$

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put $x = t$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ 0 \end{bmatrix}$$

$$= t \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

For $\lambda = 1$, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is the basis for eigenspace.

Question : 6

(a-) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$|A - \lambda I|$$

$$= \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix}$$

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$= (2-\lambda)(2-\lambda) - 1$

$= 4 - 2\lambda - 2\lambda + \lambda^2 - 1$

$= \lambda^2 - 4\lambda + 3$

$= \lambda^2 - 3\lambda - \lambda + 3$

$= \lambda(\lambda-3) - 1(\lambda-3)$

$= (\lambda-3)(\lambda-1)$

$|A - \lambda I| = 0$

$(\lambda-3)(\lambda-1) = 0$

$\lambda - 3 = 0 \qquad \lambda - 1 = 0$

$\lambda = 3 \qquad \lambda = 1$

For $\lambda = 3$ put in $(A - \lambda I)x = 0$

$$\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-x + y = 0$

$x - y = 0$

$x - y = 0 \qquad$ put $y = t$

$x = y \qquad x = t$

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$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ t \end{bmatrix}$$

$$= t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda = 1$ put in $(A - \lambda I)x = 0$

$$\begin{bmatrix} 2-1 & 1 \\ 1 & 2-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x + y = 0$$
$$x + y = 0$$

$x + y = 0$ put $y = t$

$x = -y$ $x = -t$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -t \\ t \end{bmatrix}$$

$$= t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

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(b-) $\begin{bmatrix} 2 & -3 \\ 0 & 2 \end{bmatrix}$

$A - \lambda I = \begin{bmatrix} 2 & -3 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

$= \begin{bmatrix} 2-\lambda & -3 \\ 0 & 2-\lambda \end{bmatrix}$

$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -3 \\ 0 & 2-\lambda \end{vmatrix}$

$= (2-\lambda)(2-\lambda) - 0$

$= 4 - 2\lambda - 2\lambda + \lambda^2 = 0$

$= \lambda^2 - 4\lambda + 4$ ✓

$\lambda^2 - 2\lambda - 2\lambda + 4 = 0$

$\lambda(\lambda-2) - 2(\lambda-2) = 0$

$(\lambda-2)(\lambda-2) = 0$

$(\lambda-2)^2 = 0$

$\lambda = 2, \lambda = 2$ ✓

For $\lambda = 2$ put in $(A - \lambda I)x = 0$

$= \begin{bmatrix} 2-2 & -3 \\ 0 & 2-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & -3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Date:

$-3y = 0$

$y = 0$

put $x = t$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ 0 \end{bmatrix}$$

$$= t \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(C-) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$= \begin{vmatrix} 2-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(2-\lambda) = 0$$

$$(2-\lambda)^2 = 0$$

$$2 - \lambda = 0 \qquad \lambda = 2$$

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For $\lambda = 2$ put in $(A - \lambda I)x = 0$

$$\begin{bmatrix} 2-2 & 0 \\ 0 & 2-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

put $x = t$

$y = s$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ s \end{bmatrix}$$

$$= \begin{bmatrix} t + 0s \\ 0t + s \end{bmatrix} \quad \begin{bmatrix} t \\ s \end{bmatrix}$$

$$= \begin{bmatrix} t \\ 0t \end{bmatrix} + \begin{bmatrix} 0s \\ s \end{bmatrix}$$

$$= t\begin{bmatrix} 1 \\ 0 \end{bmatrix} + s\begin{bmatrix} 0 \\ 1 \end{bmatrix} \checkmark$$

(d-) $\begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

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$$= \begin{bmatrix} 1-\lambda & 2 \\ -2 & -1-\lambda \end{bmatrix}$$

$$|A-\lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ -2 & -1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(-1-\lambda) + 4 = 0$$

$$-1-\lambda+\lambda+\lambda^2+4 = 0$$

$$\lambda^2+3 = 0$$

$$\lambda^2+3=0$$

$$\lambda^2 = -3 \rightarrow \lambda = \pm\sqrt{-3}$$

$$\lambda = \pm\sqrt{3}i \quad \lambda = \pm\sqrt{3}i$$

For $\lambda = \sqrt{3}i$ put in $(A-\lambda I)x = 0$

$$\begin{bmatrix} 1-\sqrt{3}i & 2 \\ -2 & -1-\sqrt{3}i \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(1-\sqrt{3}i)x + 2y = 0 \quad \text{---} (1)$$

$$-2x + (-1-\sqrt{3}i)y = 0 \quad \text{---} (2)$$

As eq (2)

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$(-1-\sqrt{3}i)y = 2x$

$\frac{(-1-\sqrt{3}i)}{2}y = x$

put $y = t$

$\frac{(-1-\sqrt{3}i)}{2}t = x$

$\begin{bmatrix} x \\ y \end{bmatrix} = t\begin{bmatrix} -1-\sqrt{3}i/2 \\ 1 \end{bmatrix}$

$\begin{bmatrix} x \\ y \end{bmatrix} = \left(\frac{-1-\sqrt{3}i}{2}\right)y$

For $\lambda = -\sqrt{3}i$ put in $(A-\lambda I) = 0$

Question : 7

$$\begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$|A - \lambda I| = 0$

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$$= \begin{vmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{vmatrix} - \begin{vmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix}$$

$$= \begin{bmatrix} 4-\lambda_1 & 0 & 1 \\ -2 & 1-\lambda_1 & 0 \\ -2 & 0 & 1-\lambda_1 \end{bmatrix}$$

$$|A-\lambda I| = \begin{vmatrix} 4-\lambda & 0 & 1 \\ -2 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{vmatrix}$$

$$= (4-\lambda)(1-\lambda)(1-\lambda) - 0 + 1(0-(1-\lambda)(-2))$$

$$= (4-\lambda)(1-\lambda)^2 + 1(2(1-\lambda))$$

$$= (4-\lambda)(1+\lambda^2-2\lambda) + 2 - 2\lambda$$

$$= 4+4\lambda^2-8\lambda-\lambda-\lambda^3+2\lambda^2+2-2\lambda$$

$$-\lambda^3+6\lambda^2-11\lambda+6=0$$

$$\lambda^3-6\lambda^2+11\lambda-6=0$$

The characteristic equation is

$$\lambda^3-6\lambda^2+11\lambda-6=0$$

By Synthetic Division :—

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	1	-6	11	-6
1		1	-5	6
	1	-5	6	0

$\lambda^2 - 5\lambda + 6 = 0$

$\lambda^2 - 3\lambda - 2\lambda + 6 = 0$

$\lambda(\lambda-3) - 2(\lambda-3) = 0$

$(\lambda-3)(\lambda-2) = 0$

$\lambda - 3 = 0$ $\quad$ $\lambda - 2 = 0$

$\boxed{\lambda_1 = 1}$ $\quad$ $\boxed{\lambda_2 = 3}$ $\quad$ $\boxed{\lambda_3 = 2}$

for $\lambda = 1$

$$(A - \lambda I)\vec{x} = 0$$

$$\begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$3x_1 + x_3 = 0$

$$\sim \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

For $\lambda = 1$ $\quad (A - \lambda I)\vec{x} = 0$

$$\begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_1 + R_2$

$$\begin{bmatrix} 1 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}$$

$R_2 + 2R_1$

$R_3 + 2R_1$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 + x_3 = 0$

$2x_3 = 0$ $\quad \boxed{x_3 = 0}$

$2x_3 = 0$ $\quad \boxed{x_1 = 0}$

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put $x_2 = t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

For $\lambda = 2$ $\quad (A - \lambda I)\vec{x} = 0$

$$\begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-2x_1 + x_3 = 0$ — (1)

$-2x_1 - x_2 = 0$ — (2)

$-2x_2 - x_3 = 0$ — (3)

$$\frac{1}{2}R_1 \begin{bmatrix} 1 & 0 & 1/2 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix}$$

$$\begin{matrix} R_2 + 2R_1 \\ R_3 + 2R_1 \end{matrix} \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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$$x_1 + \frac{1}{2}x_3 = 0$$

$$-x_2 + x_3 = 0$$

$$\boxed{x_3 = x_2}$$

put $x_3 = t$

$$x_1 + \frac{1}{2}t = 0$$

$$\boxed{x_1 = \frac{-1}{2}t}$$

$$\boxed{x_2 = t}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1/2\,t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -1/2 \\ 1 \\ 1 \end{bmatrix}$$

For $\lambda = 3$ $\quad (A - \lambda I)\vec{x} = 0$

$$\begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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$$\begin{matrix} \\ R_2+2R_1 \\ R_3+2R_1 \end{matrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_3 = 0 \quad \text{---(1)}$$

$$-2x_2 + 2x_3 = 0 \quad \text{---(2)}$$

$$-2(x_2 - x_3) = 0$$

$$x_2 - x_3 = 0$$

let $x_3 = t$

$$x_2 = t$$

$$x_1 + t = 0$$

$$x_1 = -t$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t \\ t \\ t \end{bmatrix} = t\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Question : B

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

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$$(A - \lambda I)$$

$$= \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 0 & -2 \\ 0 & -\lambda & 0 \\ -2 & 0 & 4-\lambda \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & -2 \\ 0 & -\lambda & 0 \\ -2 & 0 & 4-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(-\lambda)(4-\lambda) - 0 \quad -2(0-2\lambda)$$

$$-\lambda(4 - \lambda - 4\lambda + \lambda^2) - 2(-2\lambda)$$

$$-4\lambda + \lambda^2 + 4\lambda^2 - \lambda^3 + 4\lambda$$

$$-\lambda^3 + 5\lambda^2 = 0$$

$$-(\lambda^3 - 5\lambda^2) = 0$$

$$\lambda^3 - 5\lambda^2 = 0$$

The characteristic equation is

$$\lambda^3 - 5\lambda^2 = 0$$

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$$\lambda^3 - 5\lambda^2 = 0$$

$$\lambda^2(\lambda - 5) = 0$$

$\lambda^2 = 0$ $\qquad$ $\lambda - 5 = 0$

$\boxed{\lambda = 0}$ $\qquad$ $\boxed{\lambda = 5}$

$\boxed{\lambda = 0}$

For $\lambda = 0$ $\qquad (A - \lambda I)\vec{x} = 0$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 - 2x_3 = 0$

$-2x_1 + 4x_3 = 0$

put $\boxed{x_2 = t}$ $\qquad$ $\boxed{x_3 = s}$

$x_1 - 2s = 0$

$\boxed{x_1 = 2s}$

$$\begin{bmatrix} 2s \\ t \\ s \end{bmatrix} = \begin{bmatrix} 2s + 0t \\ 0s + t \\ s + 0t \end{bmatrix} = s\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + t\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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$$\lambda^3 - 5\lambda^2 = 0$$

$$\lambda^2(\lambda - 5) = 0$$

$\lambda^2 = 0$ $\qquad$ $\lambda - 5 = 0$

$\boxed{\lambda = 0}$ $\qquad$ $\boxed{\lambda = 5}$

$\boxed{\lambda = 0}$

For $\lambda = 0$ $\qquad (A - \lambda I)\vec{x} = 0$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 - 2x_3 = 0$

$-2x_1 + 4x_3 = 0$

put $\boxed{x_2 = t}$ $\qquad$ $\boxed{x_3 = s}$

$x_1 - 2s = 0$

$\boxed{x_1 = 2s}$

$$\begin{bmatrix} 2s \\ t \\ s \end{bmatrix} = \begin{bmatrix} 2s + 0t \\ 0s + t \\ s + 0t \end{bmatrix} = s\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + t\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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For $\lambda = 5$ $\quad (A - \lambda I)\vec{x} = 0$

$$\begin{bmatrix} -4 & 0 & -2 \\ 0 & -5 & 0 \\ -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-4x_1 - 2x_3 = 0$$

$$-5x_2 = 0 \Rightarrow \boxed{x_2 = 0}$$

$$-2x_1 - x_3 = 0$$

$$-2x_1 = x_3$$

put $\boxed{x_1 = s}$

$$\boxed{x_3 = -2s}$$

$$= \begin{bmatrix} s \\ 0 \\ -2s \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

Q9 - 12 are same.

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Question : 13

$$\begin{bmatrix} 3 & 0 & 0 \\ -2 & 7 & 0 \\ 4 & 8 & 1 \end{bmatrix}$$

$$(A - \lambda I)$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ -2 & 7 & 0 \\ 4 & 8 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 3-\lambda & 0 & 0 \\ -2 & 7-\lambda & 0 \\ 4 & 8 & 1-\lambda \end{bmatrix}$$

$$= (3-\lambda)(7-\lambda)(1-\lambda)$$

The characteristic equation is

$$(3-\lambda)(7-\lambda)(1-\lambda) = 0.$$